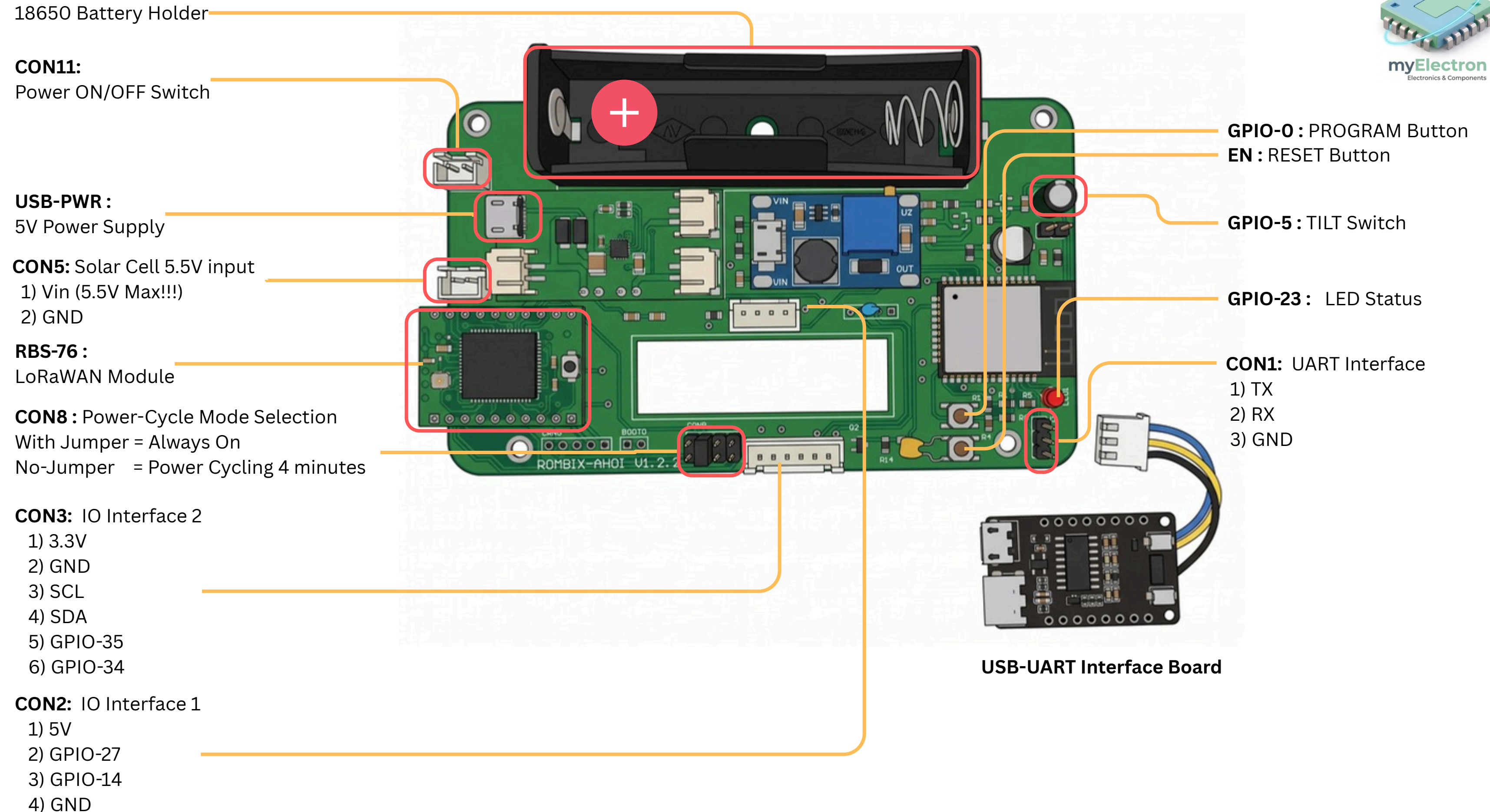


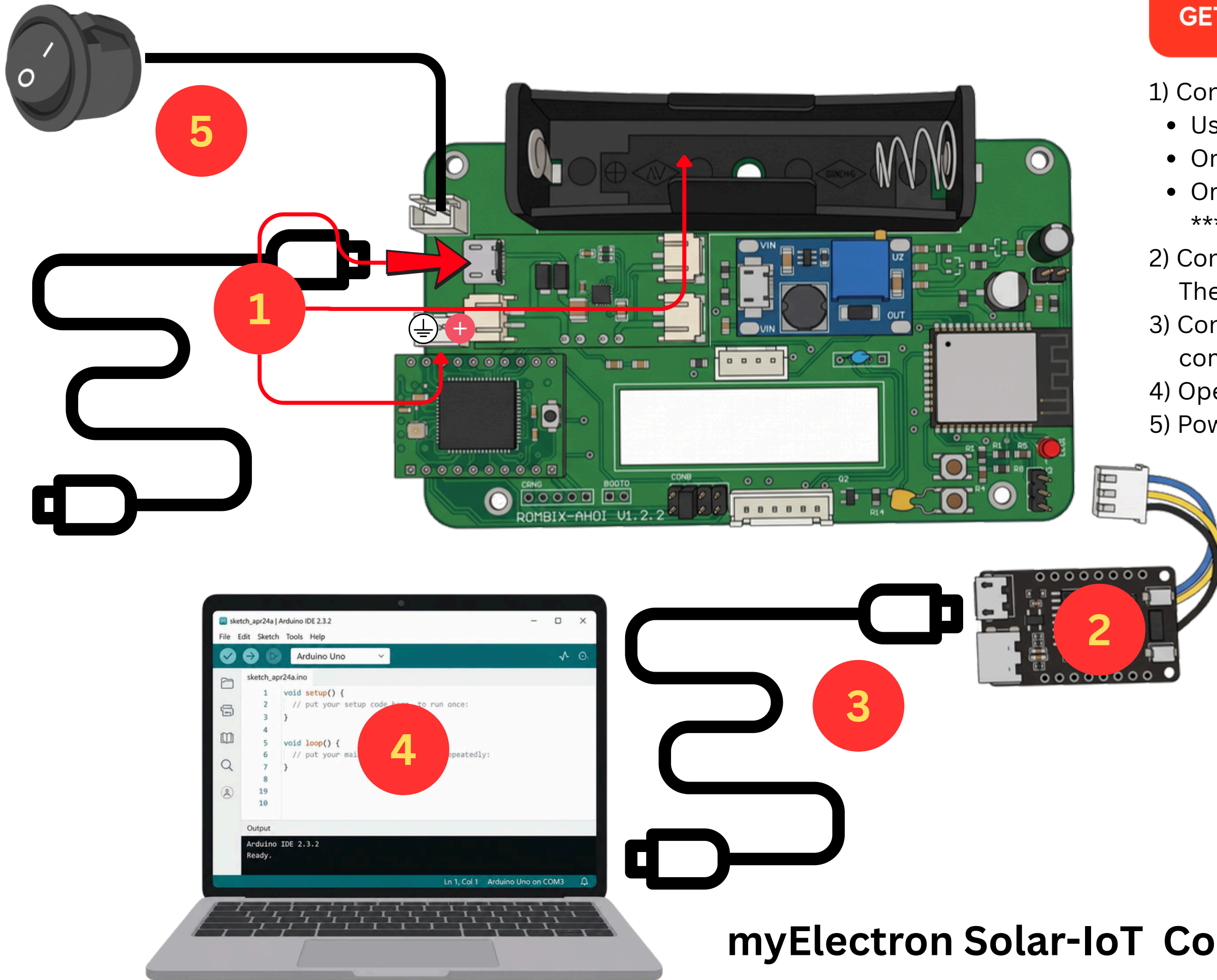


myElectron Solar-IoT Controller Pinout Diagram

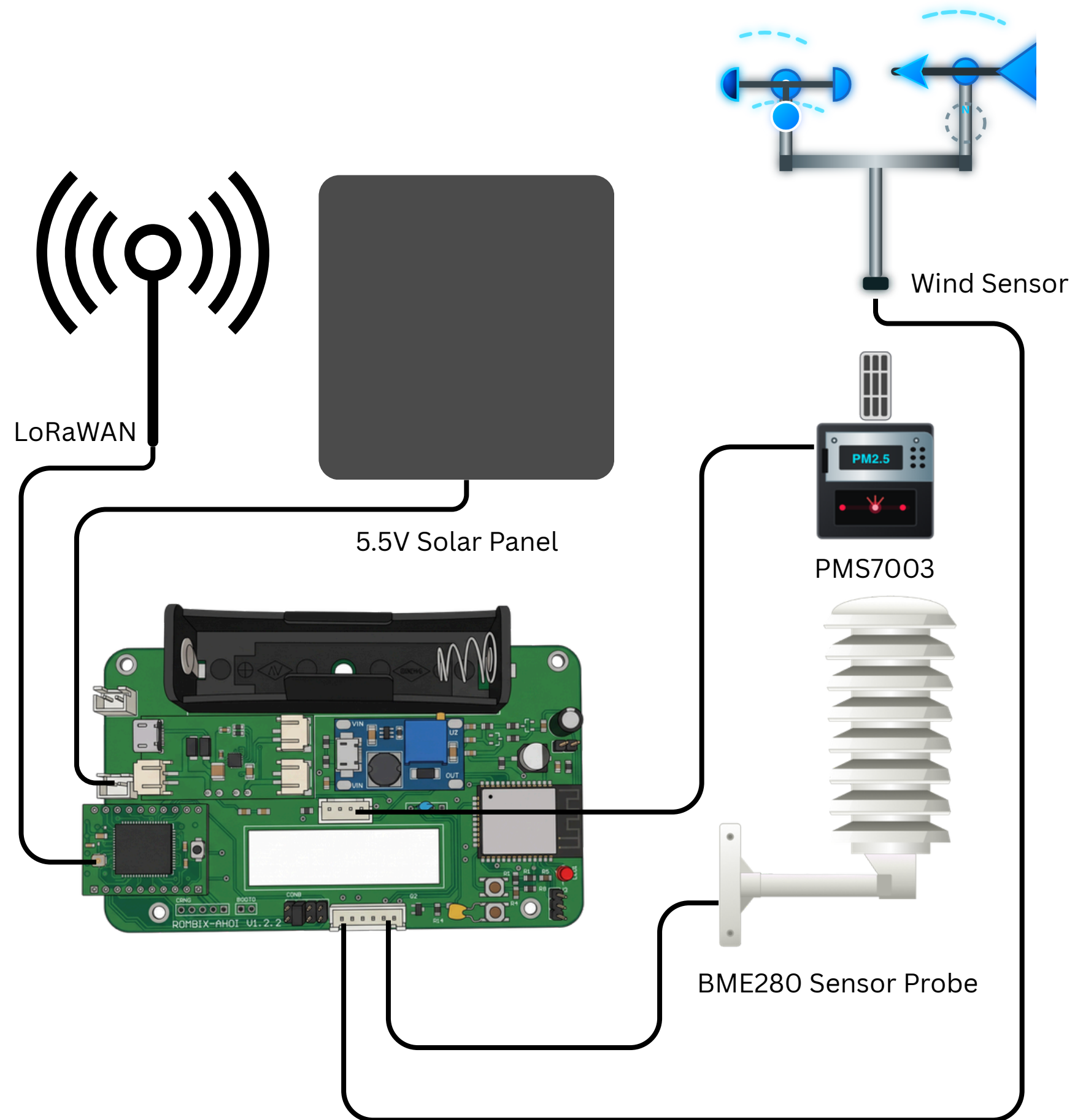
GET STARTED →

- 1) Connect 5V Power Supply,
 - Using USB power cable. (recommended 5V 2A)
 - Or insert battery Li-Ion 18650 into batt. holder.
 - Or Supply 5VDC on CON11 (Solar Cell power-port)

*** Caution : 5.5V Max. voltage ***
- 2) Connect USB-UART Interface board.
The black cable connect ground.
- 3) Connect USB Data cable to your PC and set your serial com port @115200/8/N/1
- 4) Open your serial com.port
- 5) Power on the Solar-IoT board.



myElectron Solar-IoT Controller Getting Start



What is Factory Self-Test

The Solar-IoT Controller will be preloaded at the factory with a system test program. User can record the serial number and then overwrite the existing program. The following will be tested:

- Read the chip ID (used as the board's serial number)
- Onboard battery voltage ADC
- Onboard solar PV voltage ADC
- Tilt switch status
- I2C (connected to the BME280 sensor probe)
- Pulse counter (no value, the sensor not connected)
- PMS7003 laser dust sensor (no value, the sensor not connected)
- Test communication with the LoRaWAN module



What is showing on Factory Self-Test Screen

AQI-AH01 V1.101.17

ChipID : XXXXXXXXXX *** This is your Serial No. Please noted. ***

LOOP :1

LoRa SLEEP >> sleep 900 sec uart_on

SETTING Vref...

VB = 0

OK

ADC 1425

BATT 3.66 V

Scantime = 22000

H1-max = 65535, H1-min = 65535

H2-max = 65535, H2-min = 65535

ID : XXXXXXXXXX

SN : XXXXXXXXXX

TX : 7

Init DTHP... OK!

*** Initial I2C for BME280 sensor probe ***

Vref 1895

ADC 1425

BATT 4.35 V

*** Checking battery ADC on board ***

TILT: 0

*** Checking Tilt Switch Status ***

ADC 12

PV 0.04 V

*** Checking Solar PV voltage ***

T: 30.48 °C H: 66.22 %RH P: 100876.98 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.74 Pa

T: 30.47 °C H: 66.09 %RH P: 100876.58 Pa

*** Checking BME280 ***

AN_H1_Max: 65535 AN_H1_Min: 65535 AN_H1_value: 0

AN_H2_Max: 65535 AN_H2_Min: 65535 AN_H2_value: 2245

Pulse : 1 RPM : 0.00 Speed : 0.00 m/s

Read-Time : [5219]

*** Checking Pulse counter ***

- Not connected here

Using xPM01= 0

Using xPM25= 0

Using xPM10= 0

Using Last PM1 = 0 PM2.5 = 0 PM10 = 0

*** Checking PMS7003 ***

- Not connected here

MSG = 4b01b300040be719d199ecea000.....

Initial LoRaWAN...

DEBUG-LOG >> Ok

SET ADR-OFF >> Ok

SET DR1 >> Ok

>> Ok

SET TX-CNF >> Ok

MAX-EIRP 7

>> Ok

>> Ok

DevEUI: 990990FEAA194806>> Ok

DevADDR: AA194806>> Ok

>> Ok

Join >> accepted

Sending...

....

....

....

....

LoRa SLEEP >> sleep 900 sec uart_on

>> err

Sending 3 time in 19s.

On-Time = 34 sec.

SLEEP 400 sec.

*** Checking RBS76-LoRaWAN Module ***
• Only signal test, no gateway connected.

*** Enter Deep-Sleep ***